

Intelligent Robotics : Trajectory Planning

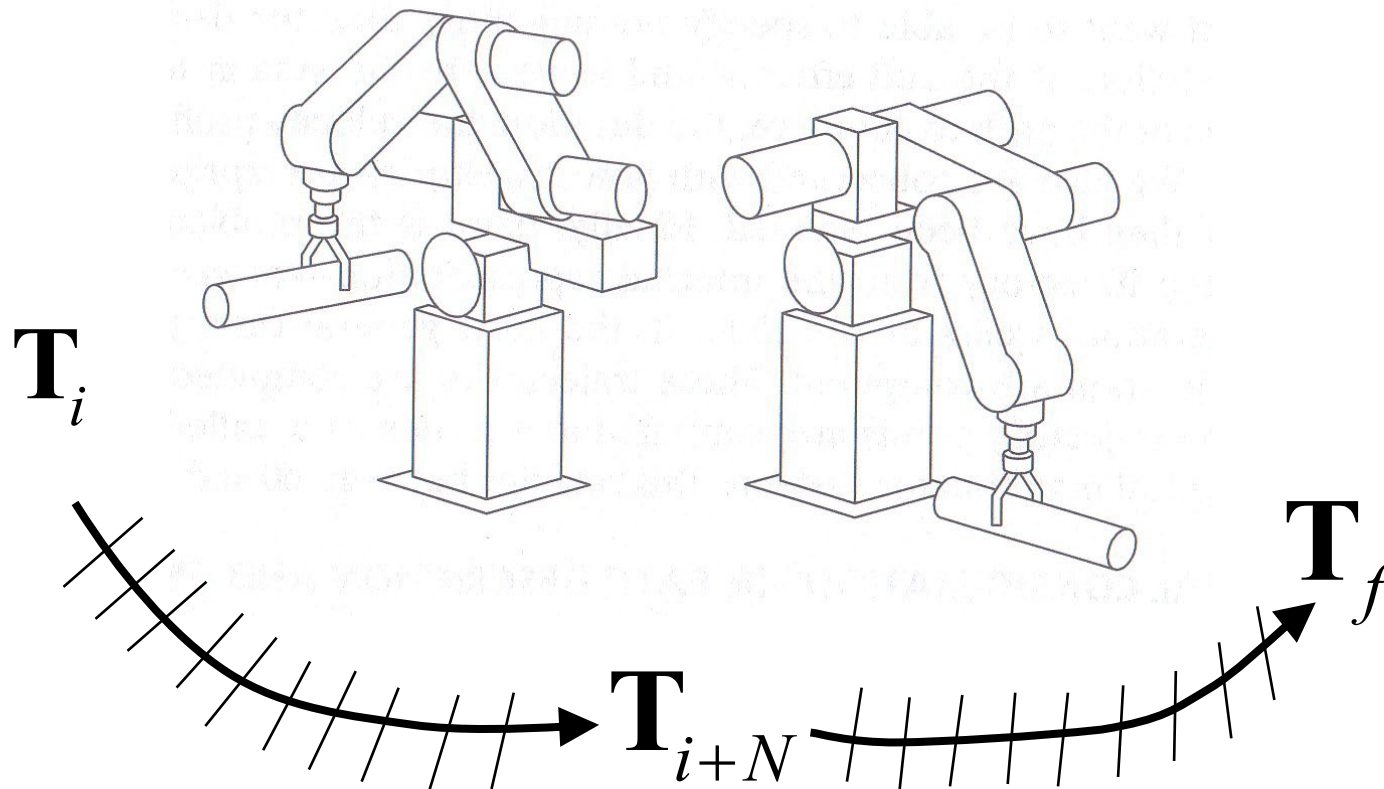
Prof. Hyungpil Moon
School of Mechanical Engineering
SungKyunKwan University

Today's Agenda

- Lecture
 - Manipulator trajectory planning
 - Mobile robot path planning

Trajectory Generation

- Let's move the manipulator from the current position(T_i) to **final** position(T_f) with **via** points.



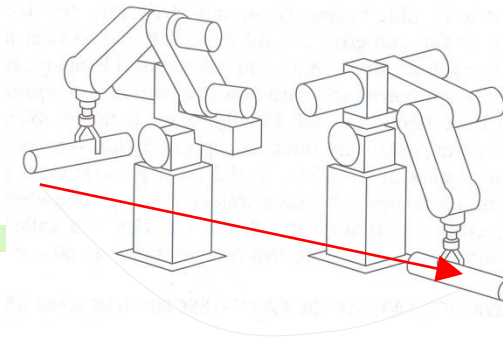
Introduction

- Trajectory: a time history of position, velocity, and acceleration for each degree of freedom
- Trajectory generation: determining a desired trajectory that passes through given initial and final points and, intermediate points
- $\mathbf{x}_i \xrightarrow{t_f} \mathbf{x}_f \rightarrow \text{inverse kinematics} \rightarrow \mathbf{q}_i, \mathbf{q}_f$
- Path update rate: the rate of trajectory computation: 60~2000Hz
- Path points = via points + initial and final points
- Trajectory planning is necessary to specify smooth temporal attributes of the motion

General Considerations

- Points $\rightarrow$ frames(position + orientation)
- Via points: intermediate points between the initial and final positions
- Path points: all the via points + initial, final points
- Smooth functions with a continuous first derivative(the second derivative is desirable)

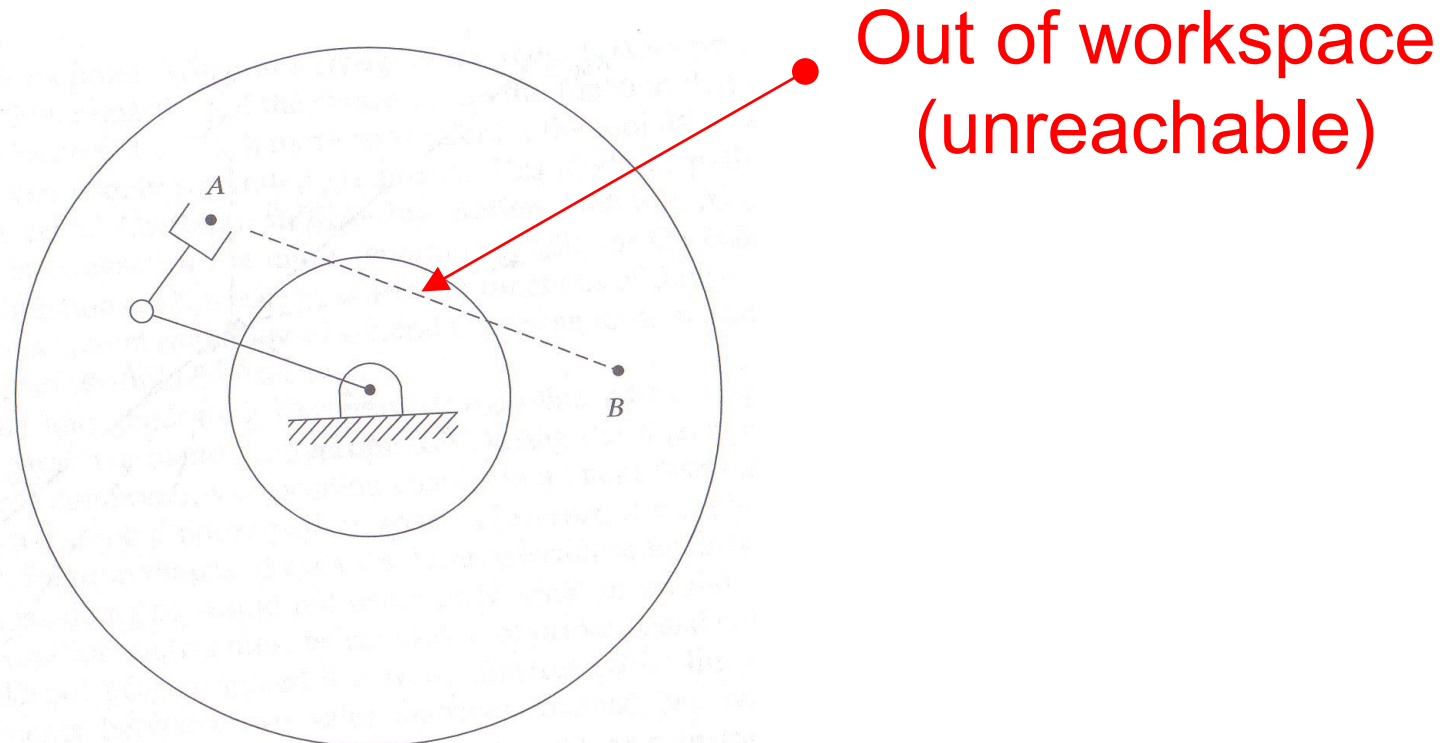
Solution Spaces



- **Joint space**
 - Easy to go through via points(solve inverse kinematics at all points and plan)
 - No problem with singularity
 - Less calculation
 - Nobody knows what happens in the intermediate points(can not follow straight line)
- **Cartesian space**
 - Physically easy to know the moving path (tracking is possible)
 - Expensive in calculation
 - Discontinuity in Cartesian space

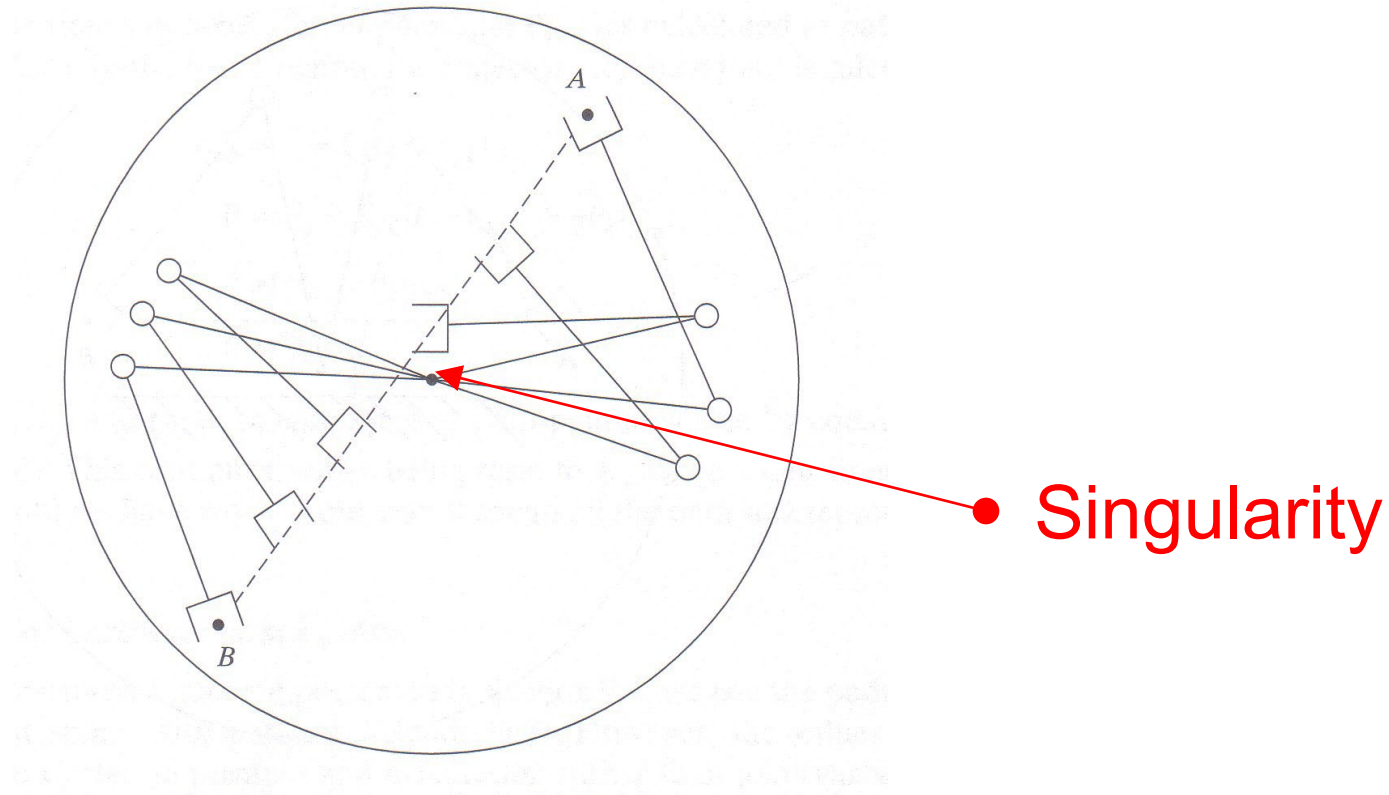
Problems with Cartesian Paths(1)

- Initial and goal points are reachable.
- Intermediate points may be unreachable.



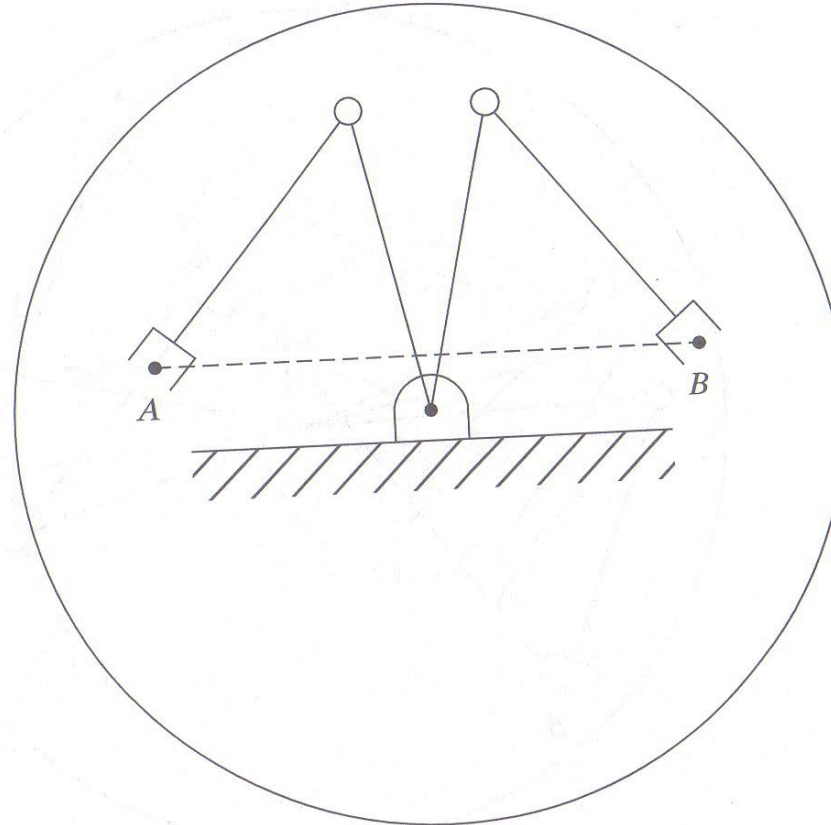
Problems with Cartesian Paths(2)

- High joint rates or velocities(infinite) near singularity.



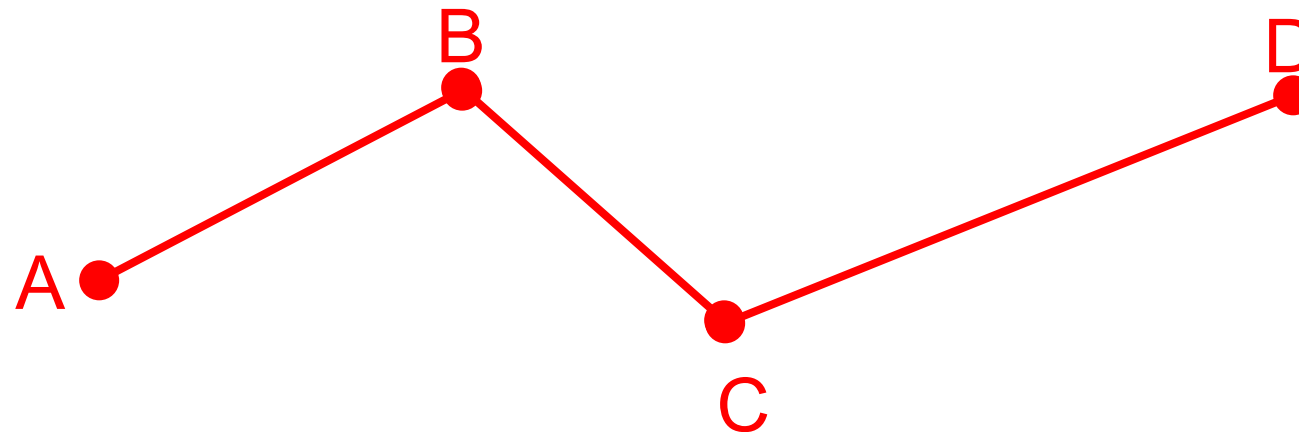
Problems with Cartesian Paths(3)

- Start and goal points are reachable in different joint space solutions



Planning in Any Space

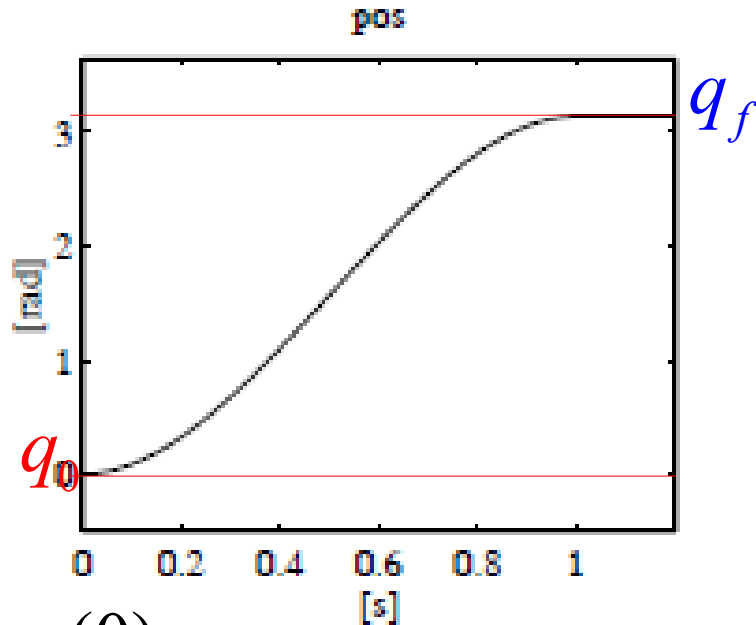
- Assume one generic variable q (it can be x, y, z or, orientation α, β, γ)
- The simplest curve pattern is:



"Discontinuous velocity at path points"

Cubic Polynomials(Splines)

$$q(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$$



$$q(0) = a_0$$

$$q(t_f) = a_3 t_f^3 + a_2 t_f^2 + a_1 t_f + a_0$$

$$\dot{q}(0) = a_1$$

$$\dot{q}(t_f) = 3a_3 t_f^2 + 2a_2 t_f + a_1$$

Initial Conditions

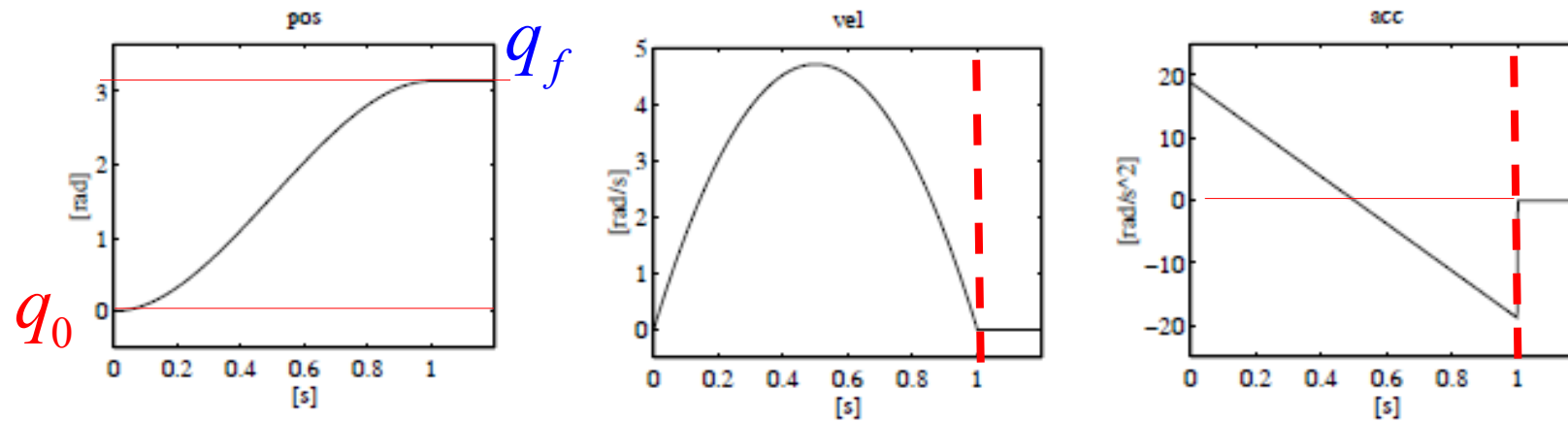
$$q(0) = q_0 \quad \dot{q}(0) = 0$$

$$q(t_f) = q_f \quad \dot{q}(t_f) = 0$$

$$a_0 = q_0, \quad a_1 = 0$$

$$a_2 = \frac{3}{t_f^2}(q_f - q_0), \quad a_3 = -\frac{2}{t_f^3}(q_f - q_0)$$

Cubic Polynomials(Splines)- (Cont'd)



Initial conditions

$$q(0) = 0 \quad \dot{q}(0) = 0$$

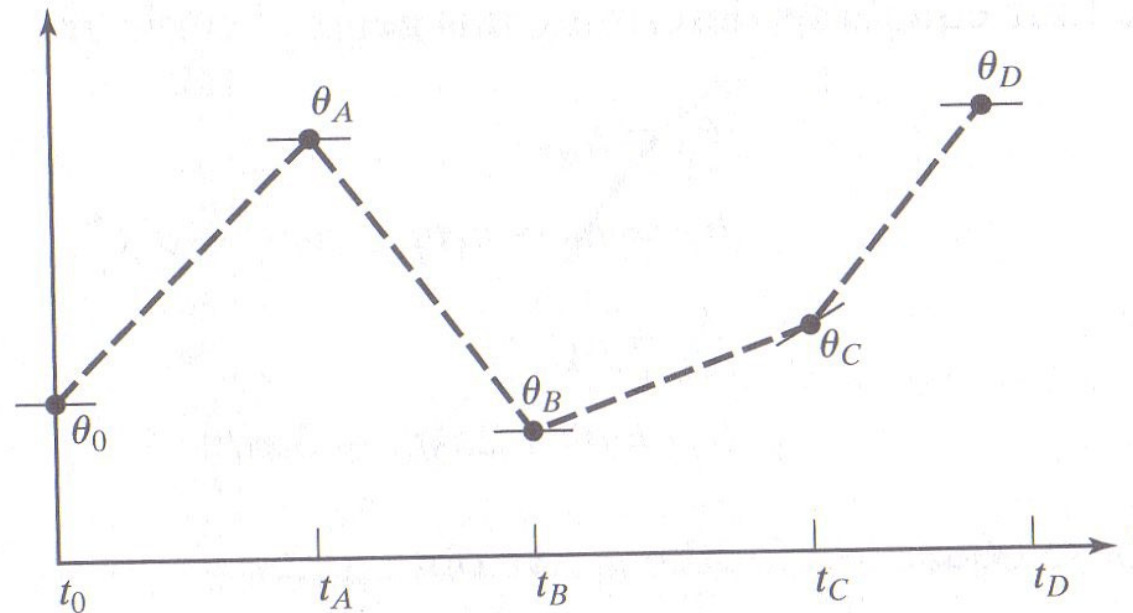
$$q(t_f) = \pi \quad \dot{q}(t_f) = 0$$

$$t_f = 1$$

$$q(t) = q_0 + \frac{3}{t_f^2}(q_f - q_0)t^2 - \frac{2}{t_f^3}(q_f - q_0)t^3$$

Cubic Polynomials with via Points(1)

- Specify intermediate via points
 - ① Stop at each point
 - Use formula from previous slides
 - ② Pass through a via point without stopping
 - Need velocities at intermediate points



Cubic Polynomials with Via Points(2)

- Changed velocity constraints
- General cubic equations

Initial Conditions

$$q(0) = q_0 \quad \dot{q}(0) = \dot{q}_0$$

$$q(t_f) = q_f \quad \dot{q}(t_f) = \dot{q}_f$$

$$q_0 = a_0$$

$$q_f = a_3 t_f^3 + a_2 t_f^2 + a_1 t_f + a_0$$

$$\dot{q}_0 = a_1$$

$$\dot{q}_f = 3a_3 t_f^2 + 2a_2 t_f + a_1$$

$$a_0 = q_0$$

$$a_1 = \dot{q}_0$$

$$a_2 = \frac{3}{t_f^2}(q_f - q_0) - \frac{2}{t_f}\dot{q}_0 - \frac{1}{t_f}\dot{q}_f$$

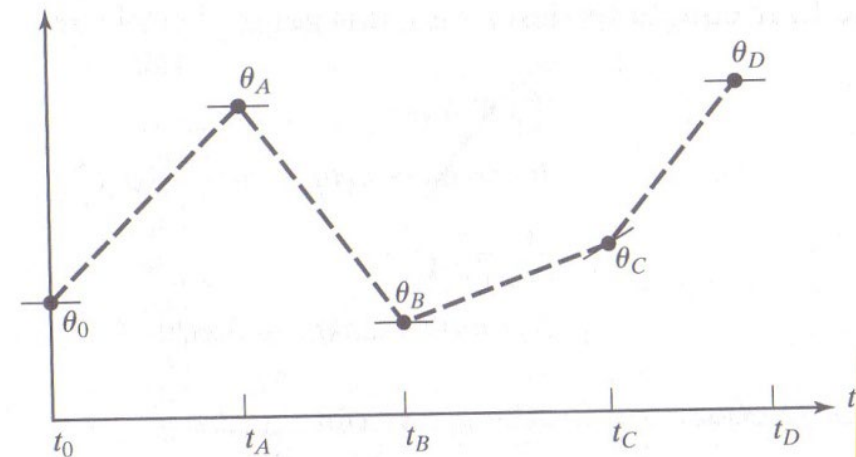
$$a_3 = -\frac{2}{t_f^3}(q_f - q_0) + \frac{1}{t_f^2}(\dot{q}_f + \dot{q}_0)$$

5th order polynomial

- Quintic Polynomial: $q(t) = a_0 + a_1t + a_2t^2 + a_3t^3 + a_4t^4 + a_5t^5$
- 6 boundary conditions
 - $q(0) = q_0, q(t_f) = q_f, \dot{q}(0) = \dot{q}_0, \dot{q}(t_f) = \dot{q}_f, \ddot{q}(0) = \ddot{q}_0, \ddot{q}(t_f) = \ddot{q}_f$
- Solutions:
 - $a_0 = q_0$
 - $a_1 = \dot{q}_0$
 - $a_2 = \frac{\ddot{q}_0}{2}$
 - $a_3 = \frac{20q_f - 20q_0 - (8\dot{q}_f + 12\dot{q}_0)t_f - (3\ddot{q}_0 - \ddot{q}_f)t_f^2}{2t_f^3}$
 - $a_4 = \frac{30q_f - 30q_0 + (14\dot{q}_f + 16\dot{q}_0)t_f + (3\ddot{q}_0 - 2\ddot{q}_f)t_f^2}{2t_f^4}$
 - $a_5 = \frac{12q_f - 12q_0 - (6\dot{q}_f + 6\dot{q}_0)t_f - (\ddot{q}_0 - \ddot{q}_f)t_f^2}{2t_f^5}$

How to Find Intermediate Velocities ?

- If we know the Cartesian linear and angular velocities, then
- $\dot{\mathbf{q}} = \mathbf{J}^{-1} \begin{bmatrix} \mathbf{V} \\ \boldsymbol{\omega} \end{bmatrix}$
- The system chooses reasonable velocities using heuristics(average of two sides etc.)
- The system chooses them for continuous velocities and acceleration such as $\dot{\mathbf{q}}_1(t_f) = \dot{\mathbf{q}}_2(0), \ddot{\mathbf{q}}_1(t_f) = \ddot{\mathbf{q}}_2(0)$



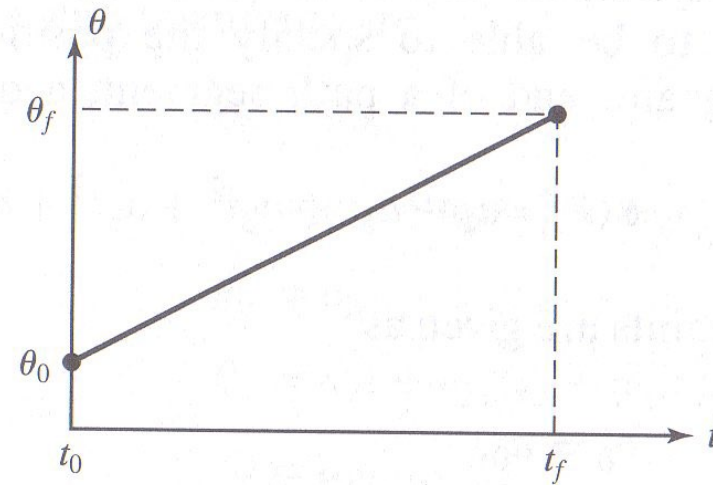
Linear Interpolation

- Straight line
- Discontinuity in velocity at the beginning and the end of the motion
- Initial conditions

$$\theta(t) = a_0 + a_1 t$$

$$\theta(t_0) = \theta_0$$

$$\theta(t_f) = \theta_f$$



Linear Interpolation with Parabolic Blends

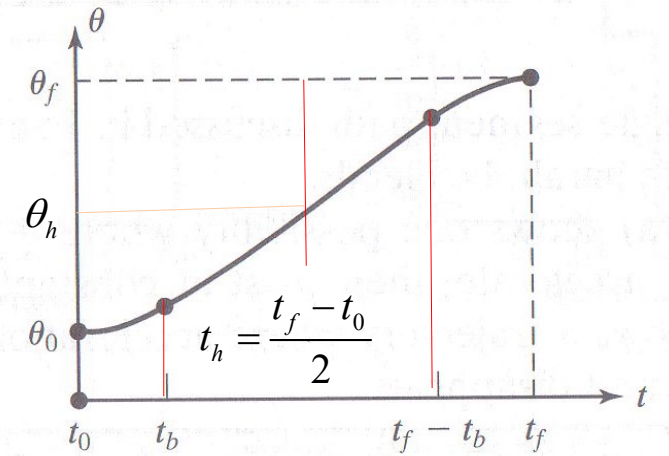
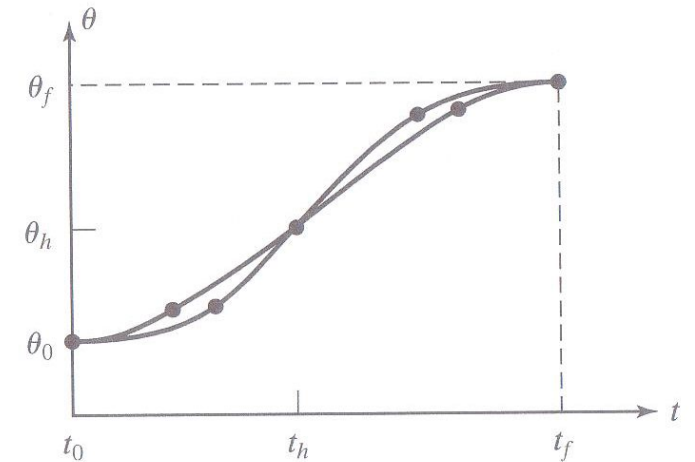
- Linear function + a parabolic blended region $\theta(t) = \frac{1}{2}\ddot{\theta}t^2 + \theta_0$
- At the blend regions
 - Constant acceleration, and linear velocity: $\ddot{\theta}(t) = a, \dot{\theta}(t) = at$,
 - Parabolic curve: multiple possible choices, but symmetric about t_h
 - Initially accelerate at constant acceleration for the blending time: $\theta_b = \theta_0 + \frac{1}{2}\ddot{\theta}t_b^2$
 - Velocity of the blending end = velocity of linear: $\ddot{\theta}t_b = \frac{\theta_h - \theta_b}{t_h - t_b}$
 - Total displacement $\theta_f - \theta_0 = \frac{1}{2}\ddot{\theta}t_b^2 + \ddot{\theta}t_b(t - 2t_b) + \frac{1}{2}\ddot{\theta}t_b^2$
- Solution model

$$\ddot{\theta}t_b^2 - \ddot{\theta}tt_b + (\theta_f - \theta_0) = 0 \quad \text{where } t = t_f - t_0 \text{ (duration time)}$$

$$t_b = \frac{t}{2} - \frac{\sqrt{\ddot{\theta}^2 t^2 - 4\ddot{\theta}(\theta_f - \theta_0)}}{2\ddot{\theta}}$$

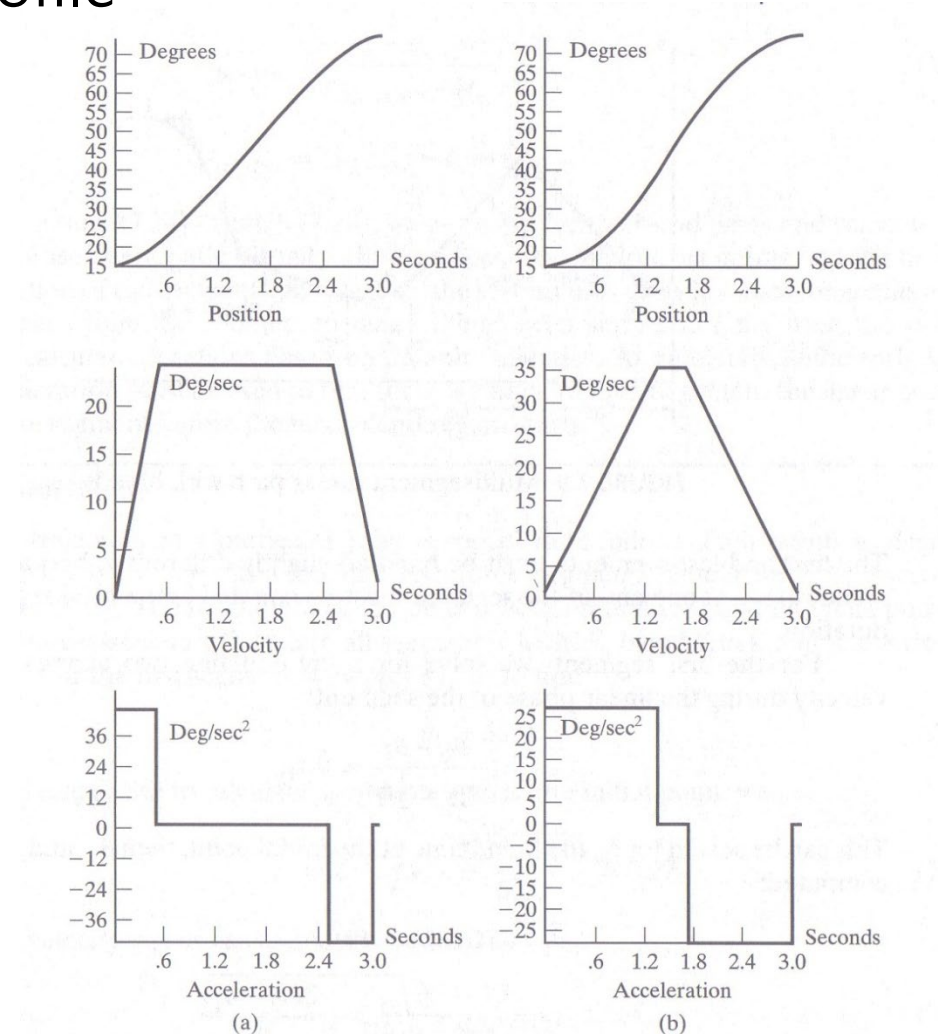
$$\ddot{\theta} \geq \frac{4(\theta_f - \theta_0)}{t^2}$$

Acceleration limit

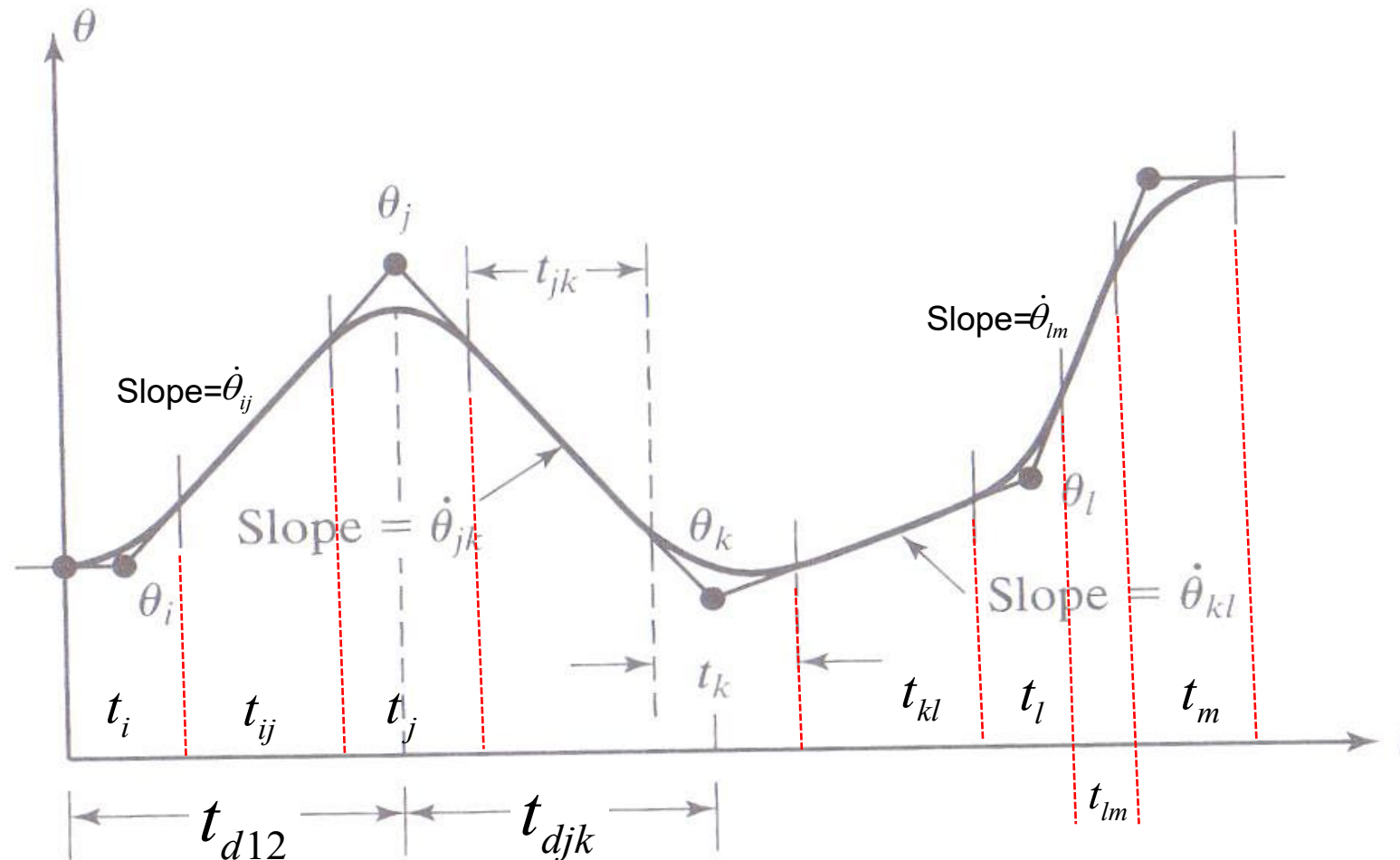


Position, Velocity, Acceleration

"Trapezoidal velocity profile"

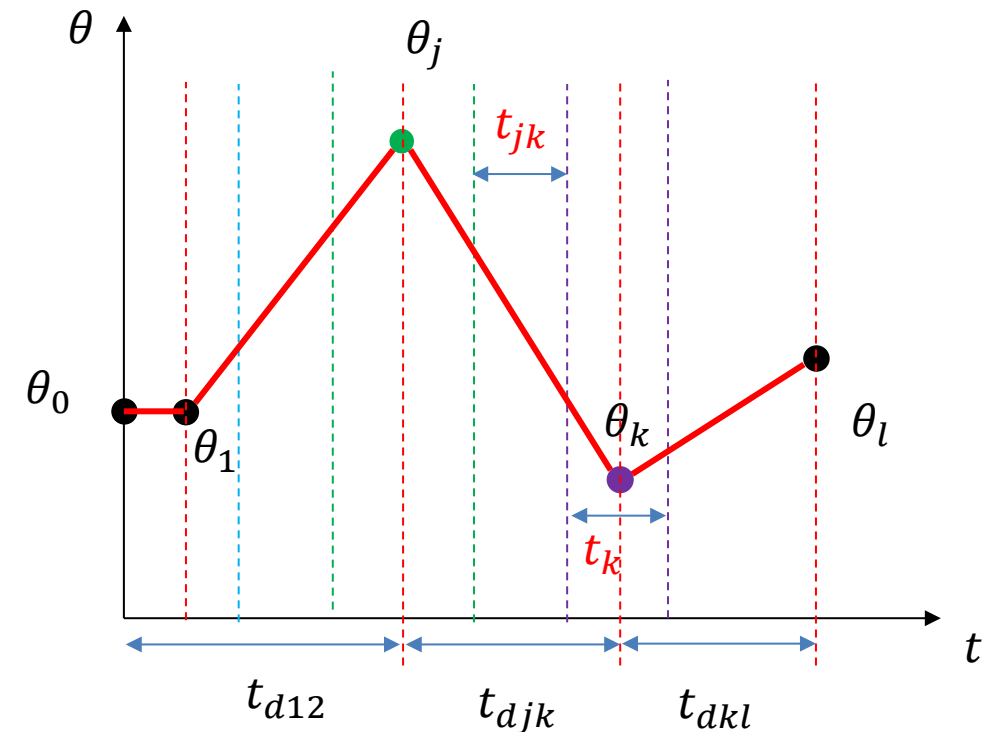


Linear Function with Parabolic Blends for a Path with Via Points



Problem Statements

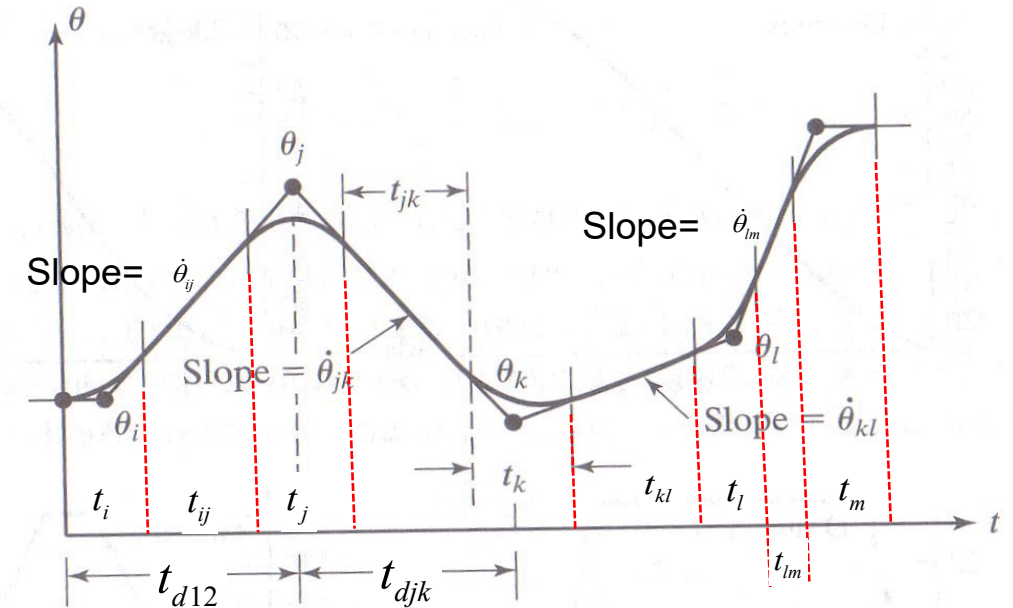
- Linear functions connect the via points, and parabolic blend are added
- Given
 - Positions: $\theta_i, \theta_j, \theta_k, \theta_l, \dots$
 - Desired overall time durations: $t_{dij}, t_{djk}, t_{dkl}, t_{dlm}$
 - The magnitude of accelerations: $|\ddot{\theta}_i|, |\ddot{\theta}_j|, |\ddot{\theta}_k|, |\ddot{\theta}_l|$
- Compute
 - Blends time: t_i, t_j, t_k, t_l, t_m
 - Straight segment time: $t_{ij}, t_{jk}, t_{kl}, t_{lm}$
 - Slopes(velocities): $\dot{\theta}_{ij}, \dot{\theta}_{jk}, \dot{\theta}_{kl}, \dot{\theta}_{lm}$
- First segment, Interior, Lat segment



- ❖ System usually calculates or uses default values for accelerations.
- ❖ The system can also calculate desired time durations based on default velocities.

Interior Path Points

- Given: $\theta_j, \theta_k, \theta_l, |\ddot{\theta}_j|, |\ddot{\theta}_k|, |\ddot{\theta}_l|, t_{djk}, \dots$
- Linear segment
 - $\dot{\theta}_{jk} = \frac{\theta_k - \theta_j}{t_{djk}}$
 - Therefore, we have also obtained $\dot{\theta}_{jk}, \dot{\theta}_{kl}$
- Sign of acceleration
 - $\ddot{\theta}_k = \text{sgn}(\dot{\theta}_{kl} - \dot{\theta}_{jk}) |\ddot{\theta}_k|$
- Blend time
 - $t_k = \frac{\dot{\theta}_{kl} - \dot{\theta}_{jk}}{\ddot{\theta}_k}$
- Duration of the linear portion between j-k
 - $t_{jk} = t_{djk} - \frac{1}{2}t_j - \frac{1}{2}t_k$



The First Segment

$$\frac{\theta_2 - \theta_1}{t_{d12} - \frac{1}{2}t_1} = \ddot{\theta}_1 t_1$$

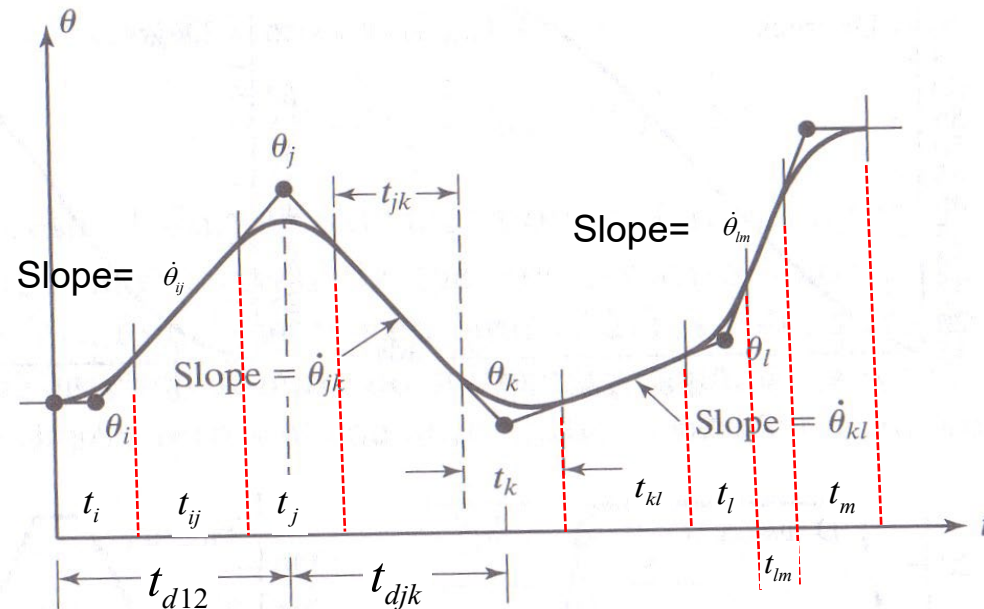
$$t_{d12} = t_1 + t_{12} + \frac{1}{2}t_2 \rightarrow t_{d12} - \frac{1}{2}t_1 = \frac{1}{2}t_1 + t_{12} + \frac{1}{2}t_2$$

$$\ddot{\theta}_1 = \text{sign}(\theta_2 - \theta_1) |\ddot{\theta}_1|$$

$$t_1 = t_{d12} - \sqrt{t_{d12}^2 - \frac{2(\theta_2 - \theta_1)}{\ddot{\theta}_1}}$$

$$\dot{\theta}_{12} = \frac{\theta_2 - \theta_1}{t_{d12} - \frac{1}{2}t_1}$$

$$t_{12} = t_{d12} - t_1 - \frac{1}{2}t_2$$



The Last Segment

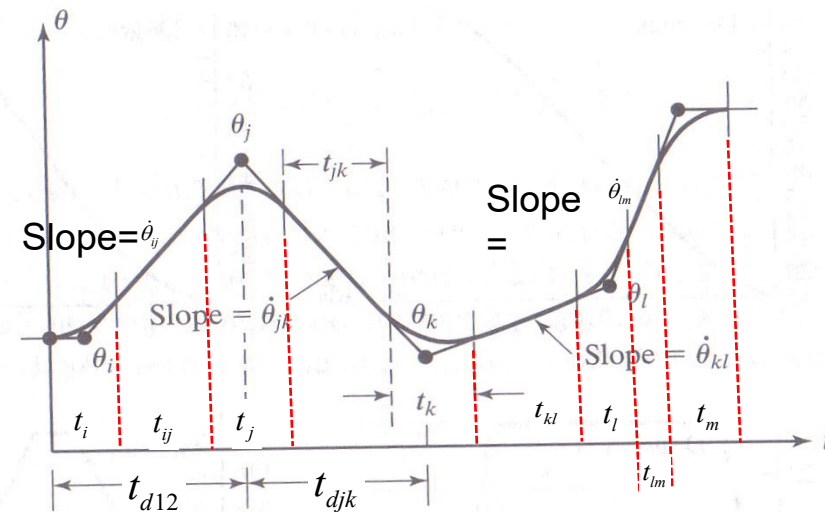
$$\frac{\theta_{n-1} - \theta_n}{t_{d(n-1)n} - \frac{1}{2}t_n} = \ddot{\theta}_n t_n$$

$$\ddot{\theta}_1 = \text{sign}(\theta_{n-1} - \theta_n) |\ddot{\theta}_n|$$

$$t_n = t_{d(n-1)n} - \sqrt{t_{d(n-1)n}^2 - \frac{2(\theta_n - \theta_{n-1})}{\ddot{\theta}_n}}$$

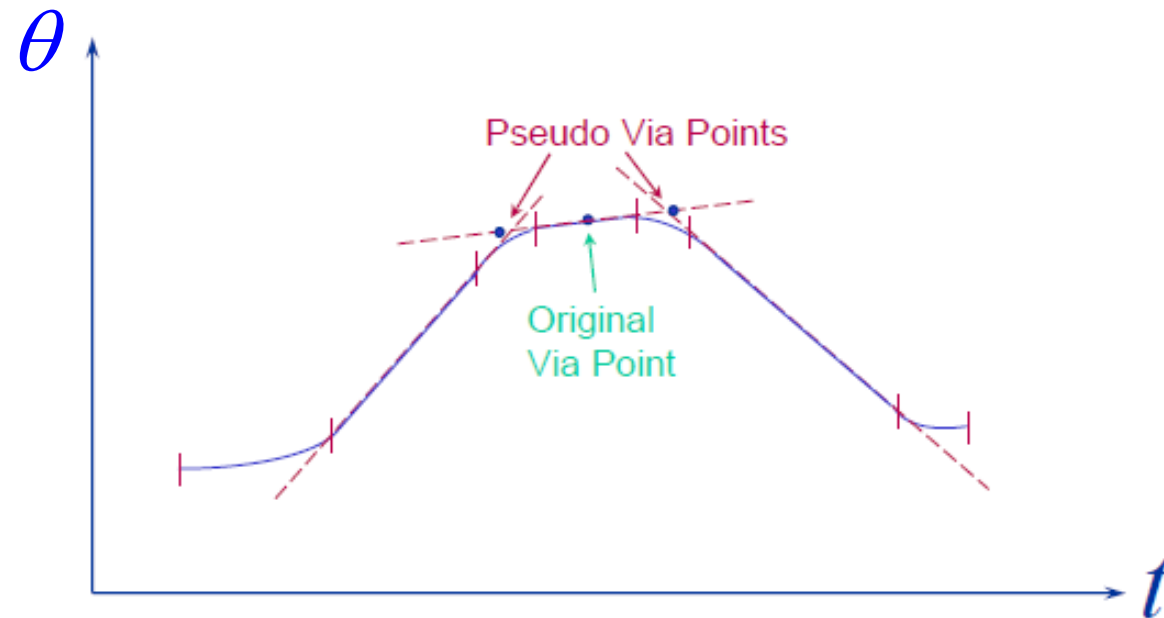
$$\dot{\theta}_{(n-1)n} = \frac{\theta_n - \theta_{n-1}}{t_{d(n-1)n} - \frac{1}{2}t_n}$$

$$t_{(n-1)n} = t_{d(n-1)n} - t_n - \frac{1}{2}t_{n-1}$$



To Go Through the Actual Via Points

- "Pseudo Via Points"
- Repeat the same procedures



Realtime Path Generation

- Trajectory in terms of $\theta, \dot{\theta}, \ddot{\theta}$ fed to the control system
 - Path generator computes at path update rate
 - In joint space directly:
 - cubic splines -- change set of coefficients at the end of each segment
 - linear with parabolic blends - check on each update if you are in linear or blend portion and use appropriate formulas for θ
 - In Cartesian space:
 - calculate Cartesian position and orientation at each update point using the same formulas
 - convert into joint space using inverse Jacobian and derivatives
- or
- find equivalent frame representation and use inverse kinematics function to find $\theta, \dot{\theta}, \ddot{\theta}$